Parts and fabrication

Everything to buy, cut and drill, drawn to scale



The finished bench, from the 3-D model (`3d/radar-bench.html`). Every size in this document comes from that model and from the horn solver, not from hand.

This document says what every part is, how big it is, and how much to buy:

- the parts list, checked against your order
- full-size cutting templates for the copper horns
- the frame, with every cut and hole
- cable runs, schematics, and the breadboard layout

Part 2, the assembly guide, uses these parts to put it together in order, with a check at the end of every stage, ending with the radar seeing someone walk across the room.

Sizes are millimetres unless marked. Print the template pages at **100 %, “actual size”**, never “fit to page”, and measure the 50 mm square on each one before cutting.

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1 The system on one page

The radar sends out a tone that sweeps steadily up from 2400 to 2483.5 MHz, then listens for echoes. An echo from something far away was sent earlier, so it comes back at a slightly lower pitch than what is being sent right now. Mixing the two together gives a low beat tone whose pitch says how far away the thing is. A sound card records that tone and the laptop turns pitch into distance.

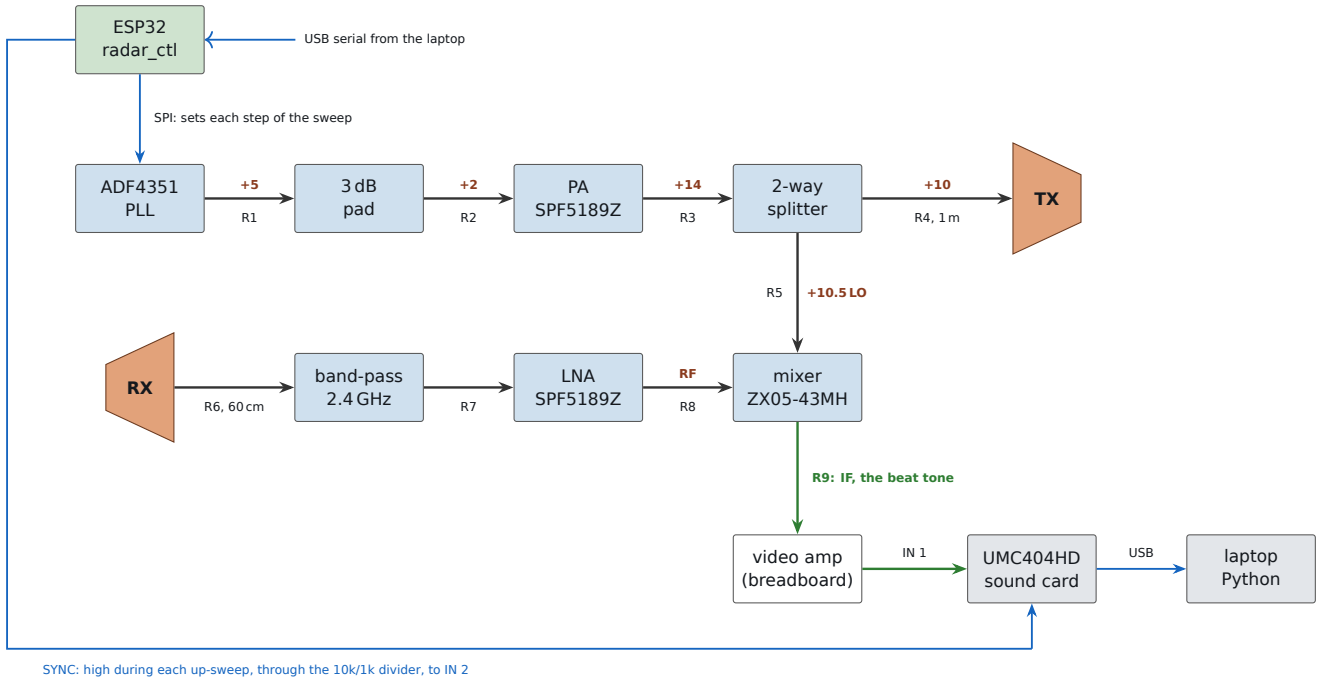


Figure 1: The signal path. Thick black arrows are coax (SMA cables); numbers in red are the signal strength in dBm at that point. Green is the low-frequency beat tone; blue is logic.

what	value
sweep	2400–2483.5 MHz, the whole 2.4 GHz ISM band
sweep timing	64 steps in 6.4 ms, repeats every 7.4 ms
distance resolution	1.80 m per range cell
speed resolution	0.13 m/s
transmit power	about +10 dBm into a 13.4 dBi horn

what	value
horns	3 copper pyramidal horns, 263.8 × 193.1 mouth
beam	about 34.2° × 36.2°
beat amplifier	gain 1111 (61 dB), cut-off 15.9 kHz
power in	12 V, 3 A supply
footprint	457 × 381 plywood, 721.9 tall

The four things you build

1. **Three copper horns** (section 3). Two face out as transmitter and receiver; the third is for the second receiver later (stage 2), and is built now so the two receivers come out as twins.
2. **The frame** (section 4). A plywood board with two uprights and two cross beams. The transmit horn sits on the top beam, the receive horn and an empty bay on the lower one.
3. **The RF chain** (section 5). Seven bought modules screwed to the board and joined with SMA cables. Nothing to make, but the cable runs matter.
4. **The breadboard** (section 7). The beat-tone amplifier, the power distribution and the logic wiring. Two op-amps and about sixty parts, no soldering.

2 What to buy

Every part the build needs. The last column is checked against your order spreadsheet (`BOM_Radar_Build.xlsx`): **21 lines are not on it**. Most of those are cheap and easy to get, or you already have them; the three you cannot work around are marked in the box below.

> **Not on your list, and the build stops without them**
>
> **The two long SMA cables** (R4, about 1 m, splitter up to the transmit horn; R6, about 60 cm, receive horn down to the band-pass filter). The 20 cm jumpers you ordered cannot reach the horns, which sit 30 and 60 cm above the board. **The ESP32 and the two TL072 op-amps**: without them nothing sweeps and nothing is amplified. Check also that you have **two** SPF5189Z modules (the PA and the LNA are the same part) and **three** copper sheets.

part	qty	what it is, what it does	your order
RF chain			
Mixer, double-balanced, SMA	1	ZX05-43MH-S+, or the generic 1.5–4.5 GHz module. IF must reach DC; LO +13 dBm	on your list
ADF4351 PLL board, SPI, SMA out	1	the sweep source, 25 MHz TCXO	on your list
SPF5189Z amplifier module	2	one is the PA, one the LNA. A 4-pack leaves spares	on your list check you have 2
2-way SMA power splitter	1	covers 2.4 GHz (e.g. 380–2500 MHz)	on your list
SMA attenuator, 3 dB	1	DC–6 GHz, between the PLL and the PA	on your list
Band-pass filter, 2.4 GHz, SMA	1	2400–2500 MHz, plain SMA (not RP-SMA)	on your list
SMA M–M jumper, RG316, 20 cm	6	runs R1–R3, R5, R7, R8 on the board	on your list check you have 6 short
SMA M–M jumper, RG316, about 1 m	1	R4: splitter up to the TX horn	**not on your list**
SMA M–M jumper, RG316, about 60 cm	1	R6: RX horn down to the band-pass	**not on your list**
SMA adapter kit (M–M, F–F barrels)	1	for whatever genders your modules have	on your list
IF pigtail: SMA male to bare leads	1	mixer IF to breadboard J1. Or cut a spare jumper in half	**not on your list**
Horns			
Copper sheet, 24 ga (0.021 in), 12 × 24 in	3	one sheet per horn, page 11	on your list check it is 3 sheets
SMA female 4-hole flange, solder cup	3	the horn feeds (plus spares)	on your list
Brass rod, 1/16 in	100 mm	three probes, cut 32 mm, trimmed to 27 mm	on your list
Plumbing solder and paste flux	1	not rosin-core electronics solder	listed, no price
Screws and nuts, M2.5 or 2-56, 8 mm	12	four per SMA flange. M3 will not fit the flange	**not on your list**
Baseband			
ESP32 devkit, 30-pin (WROOM-32)	1	runs radar_ctl: steps the PLL, drives SYNC	**not on your list**
TL072CP op-amp, DIP-8	2	the video amplifier and the half-rail reference	**not on your list**

part	qty	what it is, what it does	your order
830-point breadboard	1		**not on your list**
Resistors, capacitors, 1N5822, 3 ferrite beads	kit	full pick list on page 23	on your list
0.1 in male header strip, 40-pin	2	snapped into the 13 connectors J1–J12, JP3	**not on your list**
Solid 22 AWG jumper wire, 9 colours	48 links	the colour code on page 23	**not on your list**
USB audio interface, 4-input	1	Behringer UMC404HD: beat on INPUT 1, sync on INPUT 2	on your list
Shielded 1/4 in TS cable, 2 m	2	breadboard J2 and J12 to the UMC404HD	**not on your list**
USB cables	2	ESP32 (micro-USB) and UMC404HD (USB-B) to the laptop	**not on your list**
Power			
12 V 3 A supply, barrel plug	1	with a barrel jack to screw terminals	**not on your list**
LM2596 buck converter module	1	set to 5.00 V before it touches anything	on your list
Screw terminal block, 6-way	1	the star ground	**not on your list**
Hook-up wire, 22 AWG red and black	2 m each	5 V leads to the three RF modules	**not on your list**
Frame			
Plywood, 18 × 15 in, 3/4 in	1	the bench, page 17	listed, no price
18 mm (3/4 in) square wood	1.2 m	two uprights, 530.8 mm	**not on your list**
12 × 32 mm (1/2 × 1 1/4 in) wood strip	1.0 m	cross beams: one 444 mm, one 408 mm	**not on your list**
10 mm (3/8 in) dowel	0.5 m	two braces, about 242 mm	**not on your list**
Horn saddle clamps	3	3-D printed (PETG) or bent 3 mm aluminium, page 19	**not on your list**
L-brackets 40 mm, wood screws, M4 screws	set	uprights to plywood, beams to uprights, saddles	**not on your list**
Card and kitchen foil	1	the 293.1 × 130 mm screen behind the horns	**not on your list**
Double-sided foam tape or small brackets	1	holds the RF modules down	**not on your list**
Drone link			
RadioMaster Pocket, EdgeTX	1	the drone’s 915 MHz radio	on your list
RadioMaster Bandit Nano 915 MHz	1	in the Pocket’s bay	on your list
ELRS 915 MHz nano receiver	1	BetaFPV ELRS Nano 915 or HappyModel ES900RX	on your list

2.1 Your order, as it stands

For reference, the spreadsheet as read. Lines with no price are ones the sheet leaves blank.

line in BOM_Radar_Build.xlsx	price
ZX05-43MH-S+ mixer	$79.31
ADF4351 PLL Board	$28.32
SPF5189Z LNA	$11.81
2-way SMA Splitter	$24.56

line in BOM_Radar_Build.xlsx	price
SMA 3dB attenuator	$12.99
SMA M-M RG316	$11.97
SMA Adapters	$12.96
2.4GHz BPF	$29.10
Copper Sheet 24ga	$88.00
SMA Female 4-hole flange	$7.99
Brass Rods	$5.79
60/40 Solder + Paste Flux	—
Aviation Snips/Sheet Nibbler	—
Behringer UMC404HD	$139.00
LM2596 Buck	$7.99
Resistor + Caps	$0.00
Plywood	—
RadioMaster Pocket	$65.00
Bandit Nano 915Mhz ELRS	$40.00
BetaFPV ELRS 915Mhz Nano Rx	$10.00

2.2 Tools

tool	used for
aviation snips or a sheet nibbler	cutting the copper
steel rule, square, fine marker or scriber	marking out
60–100 W soldering iron with a big tip, or a small gas torch	the horn seams. A 25 W electronics iron cannot heat the sheet
small electronics iron, 25–40 W	header pins, the IF pigtail, the probe into the SMA cup
two lengths of angle iron or hardwood, and clamps	folding tabs on a straight line
drill with 2.7, 3.5, 4.5 and 2 mm bits; centre punch; files	connector holes, screw clearances, pilot holes, deburring
hand saw and a mitre box or square	the uprights, beams and braces
calipers (digital)	every horn size, and the probe
multimeter	seams, rails, the buck converter’s 5.00 V
8 mm (5/16 in) SMA spanner, or a small adjustable	SMA nuts: finger tight, then 1/8 turn
flush snips, wire strippers, tweezers	the breadboard
laptop with Arduino IDE (ESP32 core) and Python 3	the firmware and the radar software
gloves and safety glasses; ventilation	cut copper is sharp; flux fumes
nice to have: a VNA that reaches 2.5 GHz	measuring the horns’ match. Borrow one for twenty minutes

3 The copper horns

Each horn is a rectangular pipe with a flared mouth, made from nine flat pieces of 24 gauge copper soldered along folded tabs. The pipe is 86.4 × 43.2 inside and 115 long; the flare opens to 263.8 × 193.1 over 91.5. A brass rod pushed through the pipe wall from an SMA connector is the feed. Make three, identical.

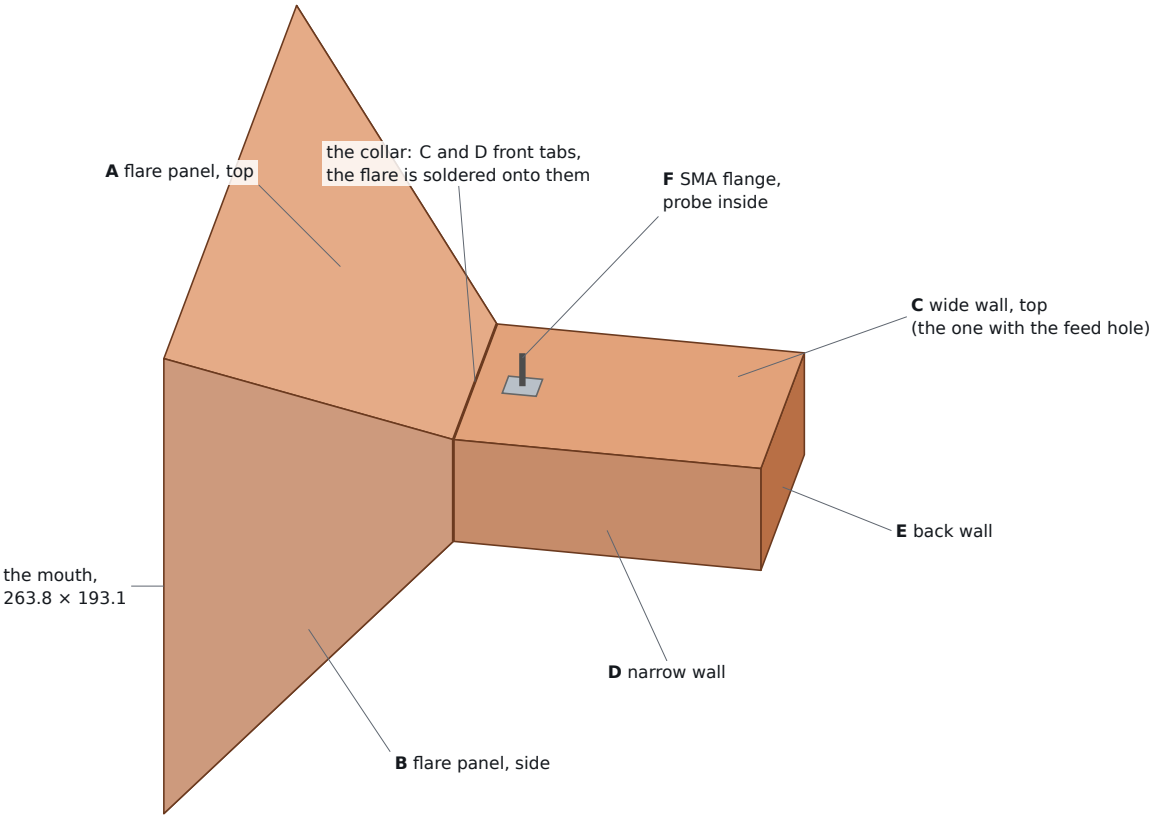


Figure 2: One horn seen from behind and above, as it comes off the bench before it is rolled onto its side for mounting. There are two each of A, B, C and D, and one E. The bottom C has no hole.

piece	name	qty	the wall itself (tabs extra)	folds
A	flare panel, top and bottom	2	trapezoid, edges 86.4 and 263.8, 118.3 high	none
B	flare panel, sides	2	trapezoid, edges 43.2 and 193.1, 127.5 high; tabs on both slanted edges	64°
C	pipe wall, wide	2	87.5 × 115; tabs on both long edges and the front	90° sides, 39° front
D	pipe wall, narrow	2	43.2 × 115; tab on the front edge	44° front
E	back wall	1	88.5 × 44.3; tabs on all four edges, corners notched	90°
F	SMA flange and probe	1	bought; brass rod cut 32, trimmed to 27 inside	—

Every tab is 6 mm wide. The slanted edges of A and B must both come out 147.9 mm, because that is where they meet; if they differ the flare will not close. C and E are cut a little over the pipe's 86.4 because the sheet is 0.53 mm thick and they lap over the other walls: the inside still comes out 86.4 × 43.2.

copper, outside face up tab fold line centre line drill

3.1 The flare panels, flat

Drawn to scale (1 : 2), each as seen from its outside face. The long edge is the mouth; the short edge meets the pipe.

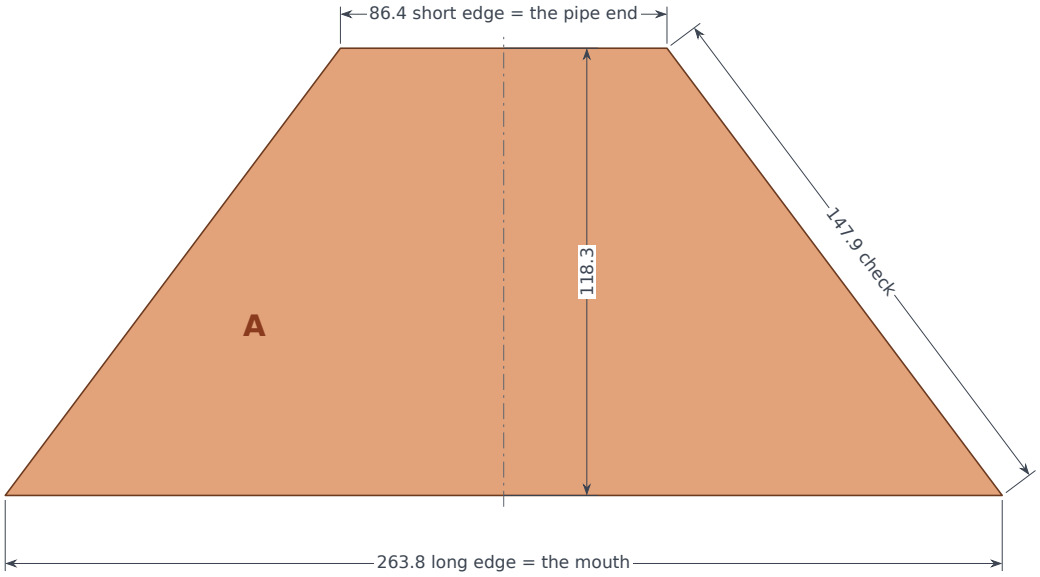


Figure 3: Piece A, top and bottom flare panel. Cut 2. No tabs: the B tabs lap over its slanted edges.

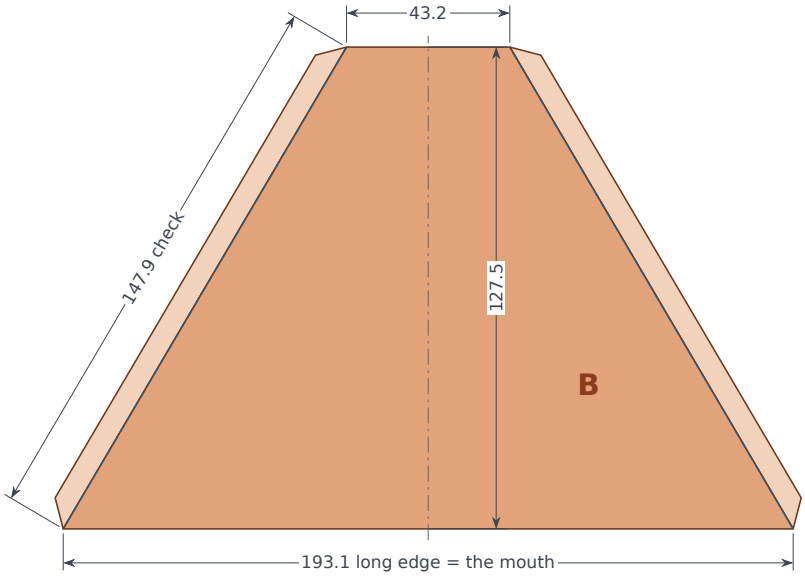


Figure 4: Piece B, side flare panel. Cut 2. The tabs on its slanted edges fold 64° and wrap over the outside of A.

3.2 The pipe pieces, flat

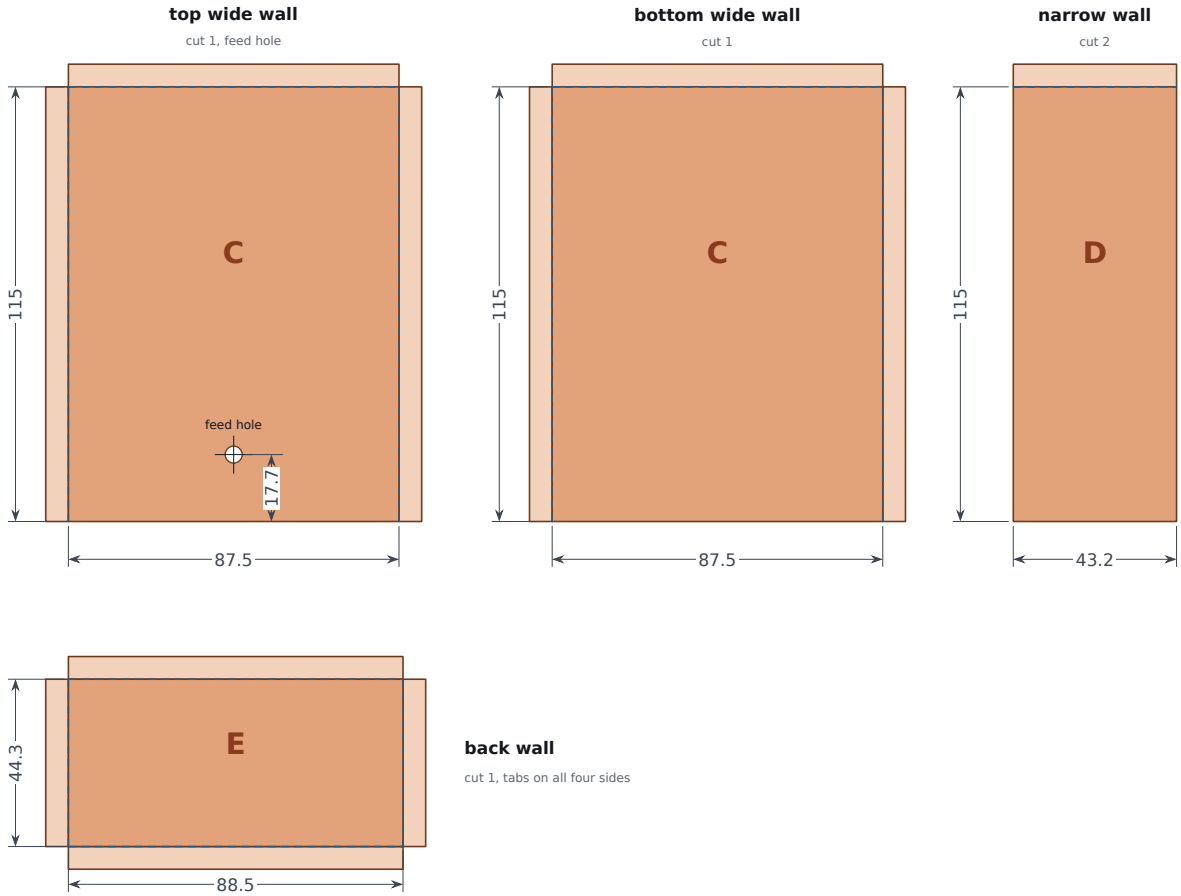


Figure 5: The pipe pieces at 1 : 2, back edge at the bottom. Only one C gets the feed hole: 17.7 mm up from the back edge, halfway across. The strips above C and D are the front tabs that become the collar the flare sits on.

3.3 The feed: connector and probe

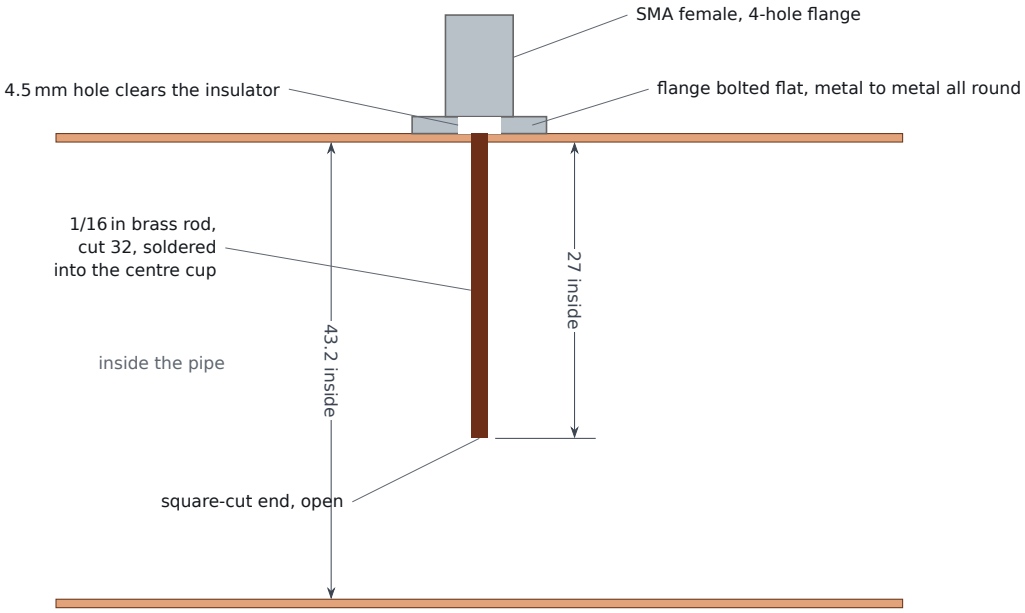


Figure 6: Section through the top wide wall at the feed, full height of the pipe shown. The flange's four 2.7 mm holes take M2.5 or 2-56 screws (M3 is too big).

3.4 One horn per sheet

Everything for one horn fits on one 12 × 24 in sheet with a 3 mm cutting gap between pieces and 4 mm clear of the edge. Each horn uses about 178 square inches, 62 % of a sheet. Two sheets do *not* make three horns: what is left over is too narrow for another set of flare panels. **Three horns need three sheets**; a fourth is cheap insurance against a mis-cut.

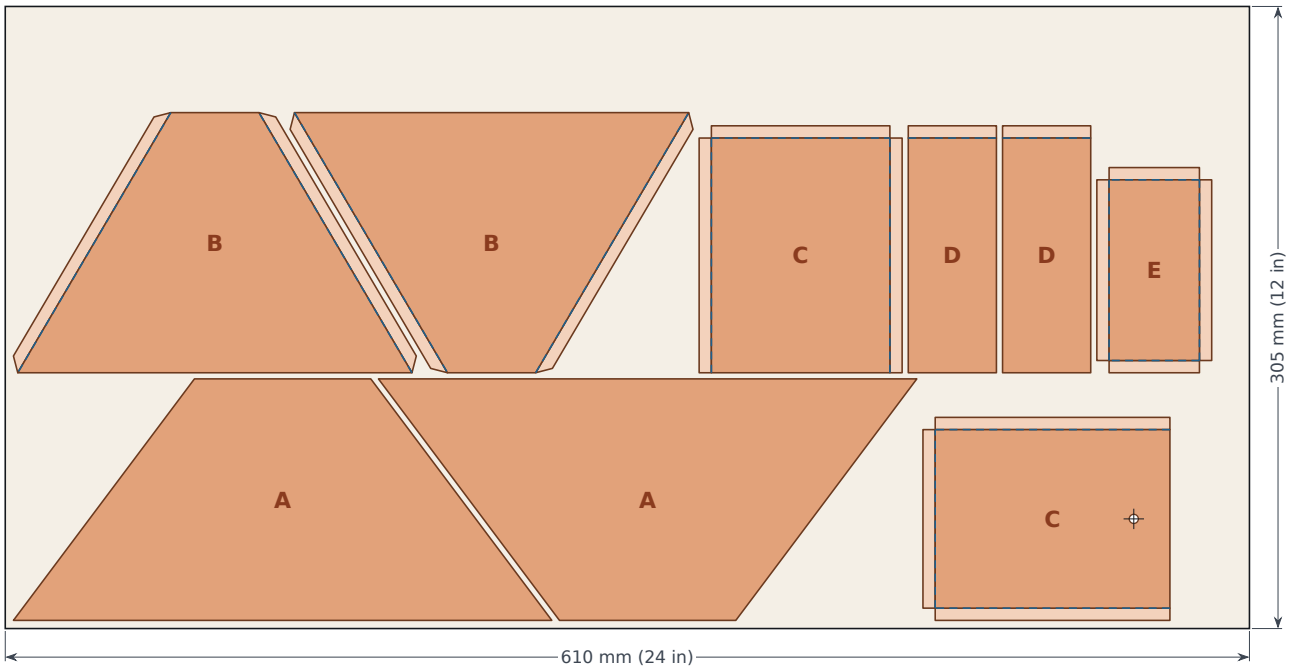


Figure 7: One horn’s nine pieces on one sheet, to scale. Turn every second trapezoid round so the pair nest. The spare strip along the top is for practising a fold and a soldered seam first.

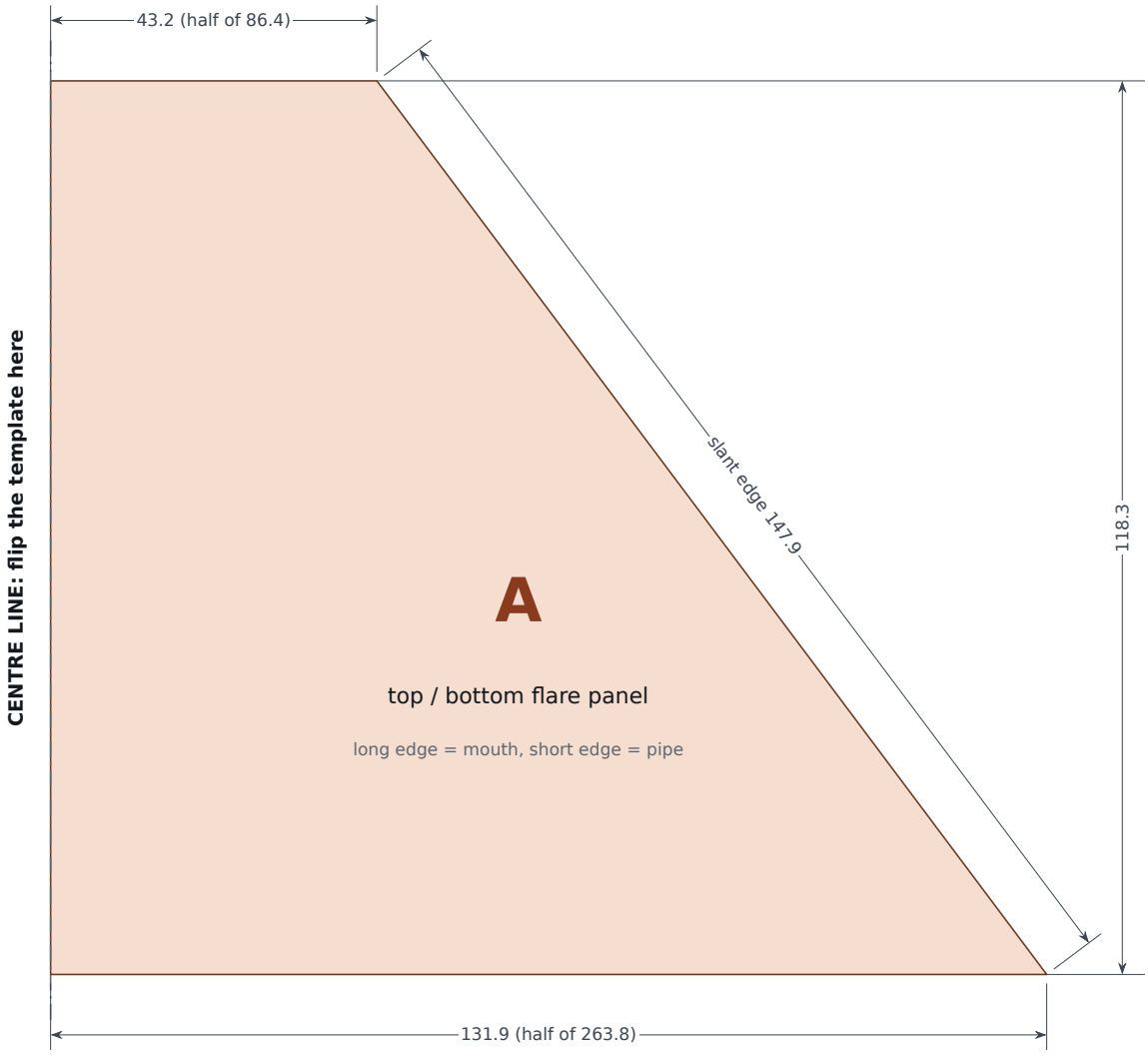
Marking a trapezoid accurately

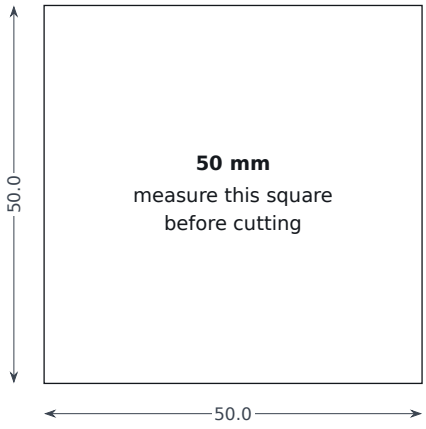
Never mark the slanted edges by measuring along them. Draw the two parallel edges and a centre line square to them, measure half the width out each side of the centre line on each parallel, then join the points. The slant comes out right by itself; check it with the rule against 147.9. The full-size templates on the next pages are half-templates for exactly this reason: trace one side, flip the template over on the centre line, trace the other.

> **Cutting on a waterjet instead**
>
> `build/waterjet/` has the cut file for an OMAX abrasive waterjet: this sheet as a DXF, the nine true-size outlines plus the feed hole, spaced 5 mm apart for the jet. Cut its 50 mm test coupon first. Its README walks through LAYOUT and MAKE, holding thin copper down, and what to do after. The templates below are then only for checking parts.

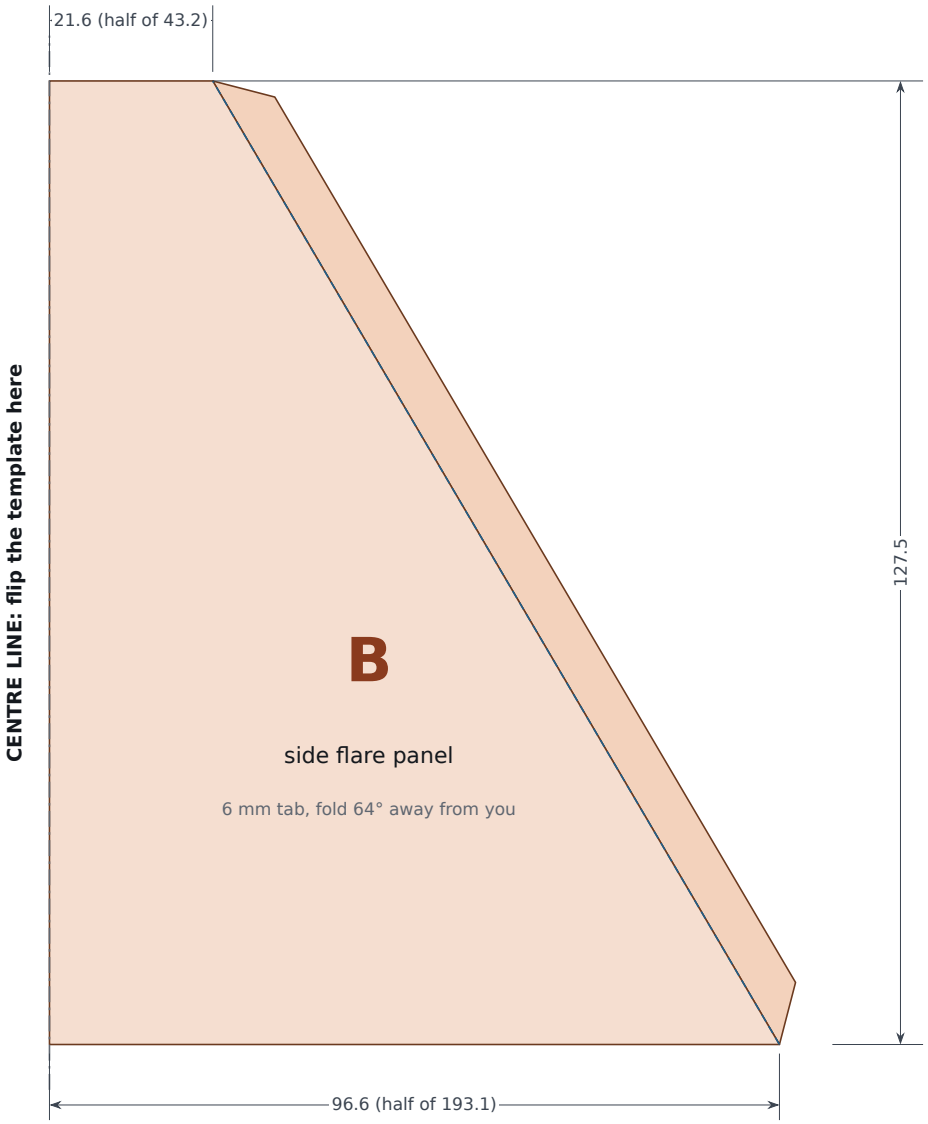
> **Using the full-size templates**
>
> Print pages 12–14 at **100 % / actual size**. Measure the 50 mm square on each page; if it is not 50.0 mm, your printer scaled it and the cut will be wrong. Cut the paper out, tape it to the copper with the outside face up, and scribe round it. The numbers on the drawing win over the paper if they disagree.

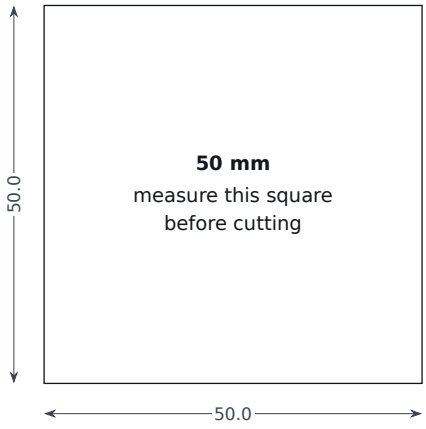
3.5 Template: flare panel A (full size)





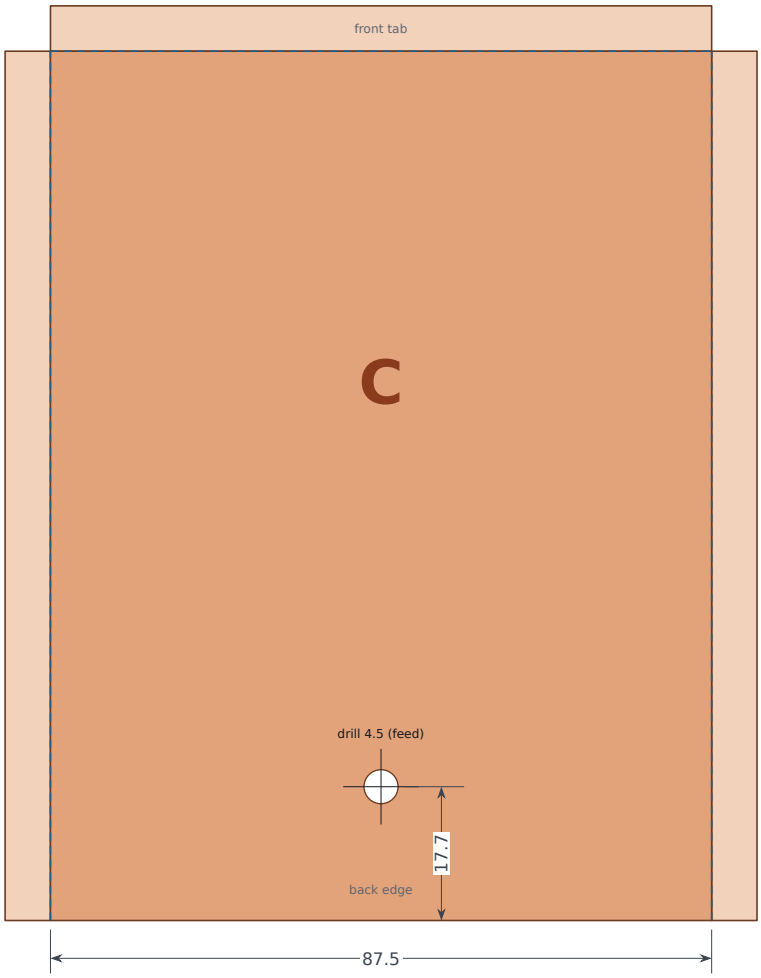
3.6 Template: flare panel B (full size)



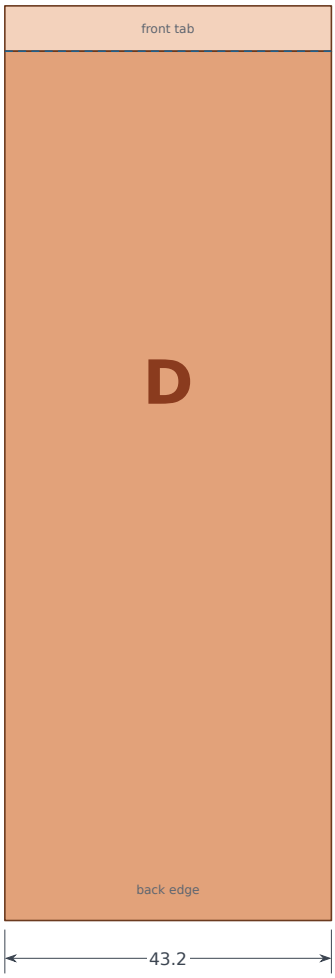


3.7 Template: the pipe, C, D and E (full size)

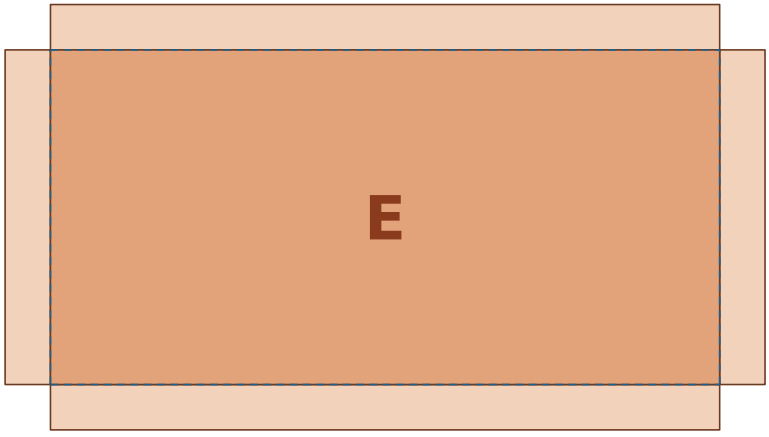
wide wall: cut 2, feed hole in one only

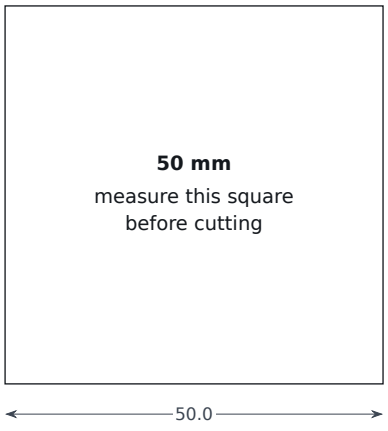


narrow wall: cut 2



back wall: cut 1, tabs on all four sides





4 The frame

A plywood board, two square uprights, two cross beams and two diagonal braces. Wood is easiest; aluminium bar of the same section works the same way. Horns are *rolled* 90° on their side: the pipe's wide walls face left and right and the long side of the mouth runs up and down. Roll all three the same way, so every connector points to the left as you sit behind the bench. (Left and right in this document are always as seen from there, the operator's seat, with the horns pointing away from you into the room.) That is what the beam heights below assume, and it makes the two receive cables follow the same path when the second receiver goes in.

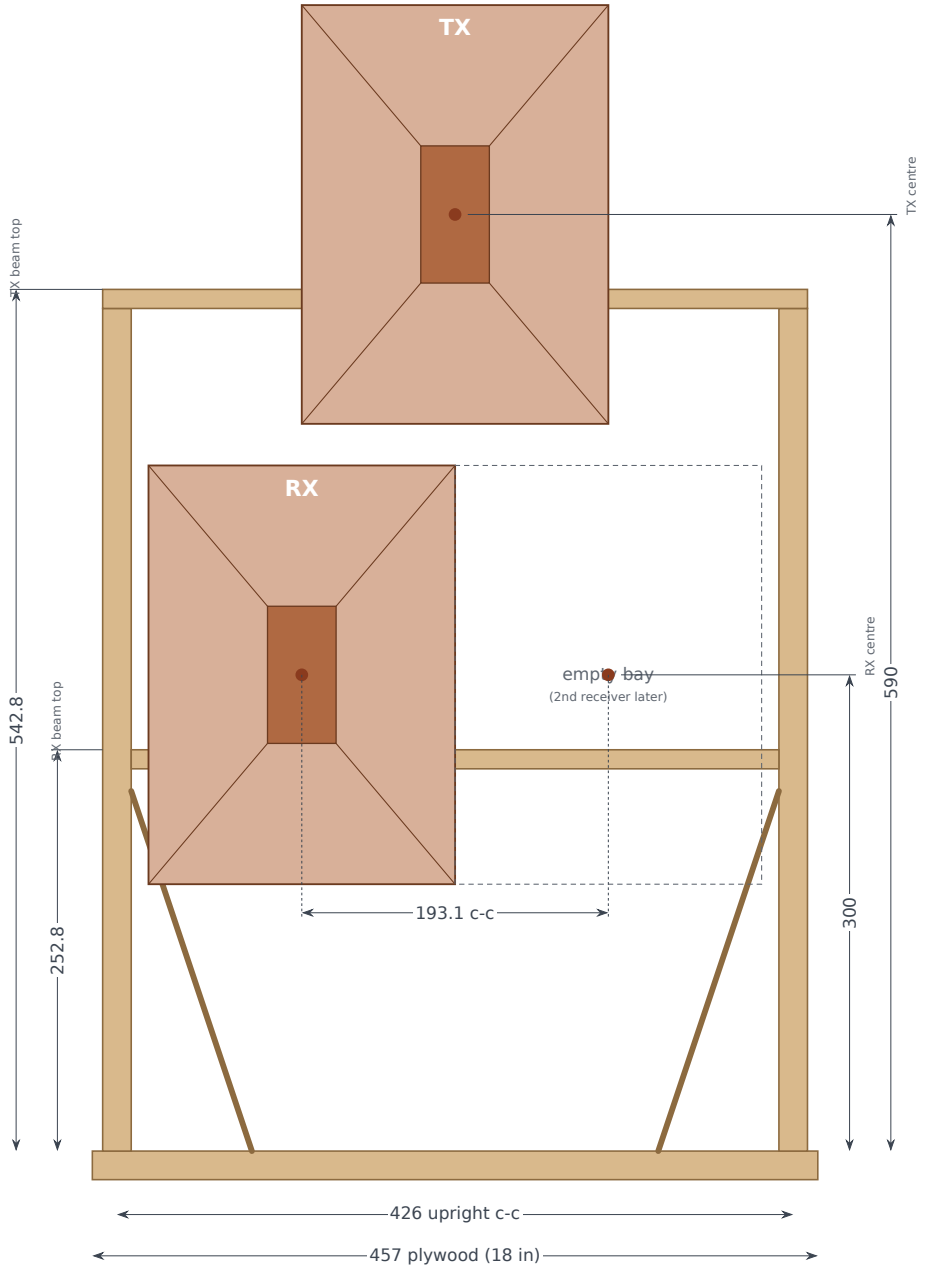


Figure 8: The frame from behind, from where you sit, 1 : 4; the horns point away from you. The receive horn and the empty bay sit side by side on the lower beam, mouths touching; the transmit horn sits centred on the top beam, 26.2 mm clear of them. Red dots are the horn centre lines. Braces run from the uprights down to the board 128 mm either side of centre.

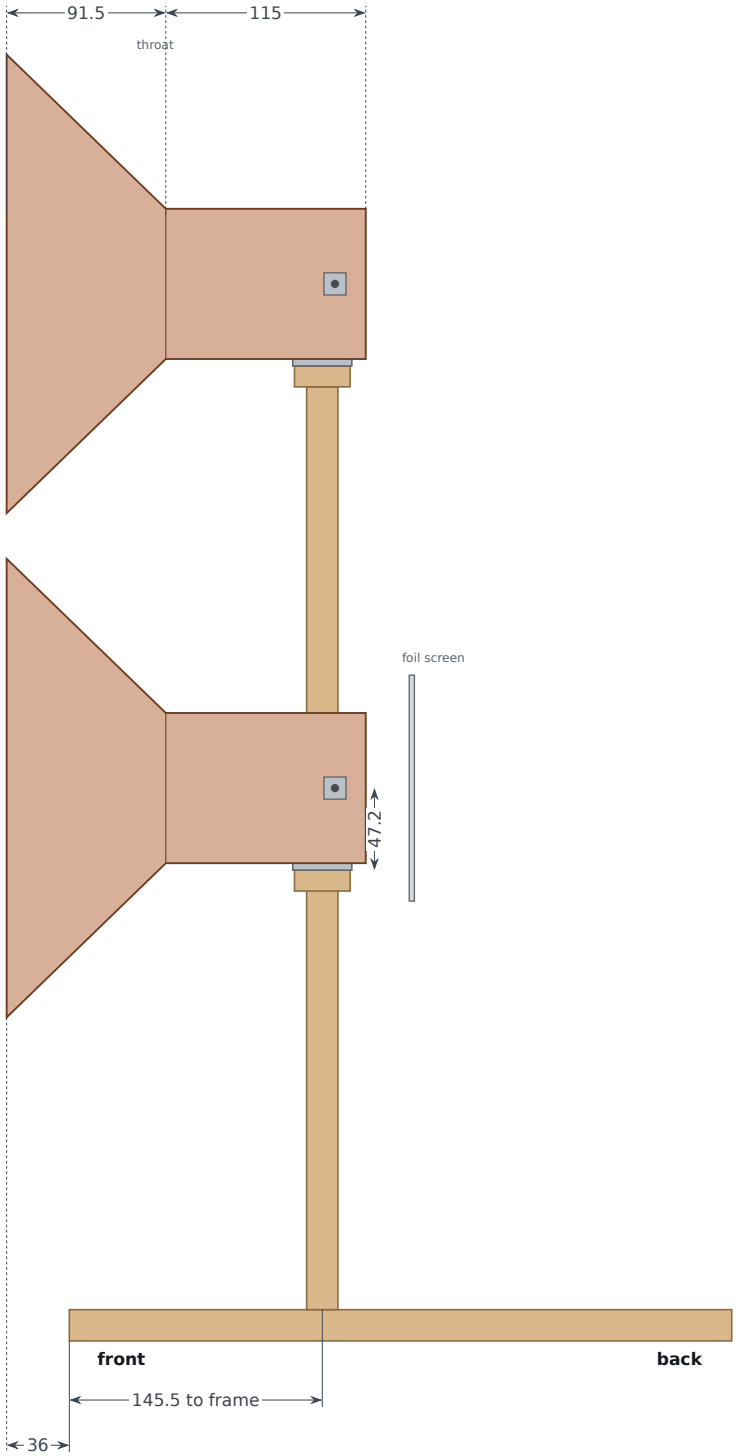


Figure 9: From the left side, 1 : 4; the horns point left in this view. The horns hang forward of the frame, each held by a saddle clamp on its pipe. The mouths are flush with each other and overhang the front edge of the board by 36 mm. A foil screen 50 mm behind the frame stops the board’s electronics from seeing the backs of the horns.

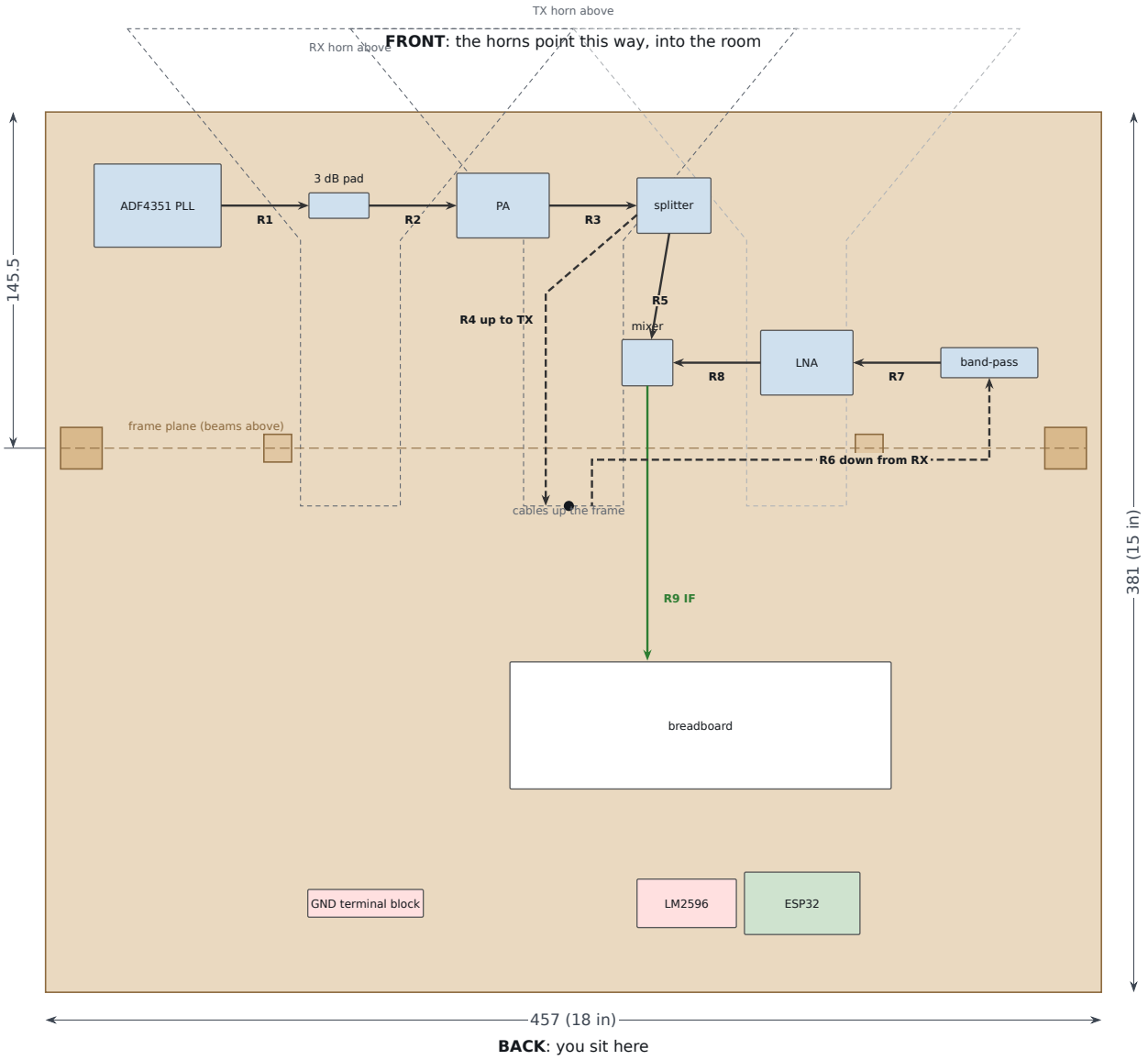


Figure 10: Top view of the board, 1 : 3: horns at the top, your seat at the bottom. Dashed outlines are the horns overhead. R1–R8 are the short SMA jumpers; R4 and R6 are the long runs to the horns.

4.1 Cut list

part	qty	stock	cut and drill
base board	1	3/4 in plywood	457 × 381 (18 × 15 in). Mark the centre line front to back
upright	2	18 mm (3/4 in) square	530.8 long, square ends. Pilot hole in the top end for the TX beam screw; one 3.5 mm hole through at the RX beam
TX beam (top)	1	12 × 32 mm strip	444 long. Two 3.5 mm holes 426 apart, centred; the TX saddle in the middle
RX beam (lower)	1	12 × 32 mm strip	408 long, fits *between* the uprights. Pilot holes in both ends; saddles at ± 96.55 from centre
brace	2	10 mm dowel	about 242.2 long; both ends cut at 69° to the length (cut to fit)
saddle clamp	3	PETG print, or 3 mm aluminium	page 19
foil screen	1	card + kitchen foil	293.1 × 130, foil side toward the horns

Why the beams join the way they do

The 3-D model lets the beams pass through the uprights. Real wood cannot, so: the **TX beam sits on top** of both uprights, one screw down into each end; the **RX beam fits between them**, one screw in through each upright into its end. Both beam tops land exactly where the model puts them, 252.8 and 542.8 above the board. Each upright foot gets **two** L-brackets, one on its front face and one on its back, because the horns hang forward and pull the frame that way.

4.2 Part drawings

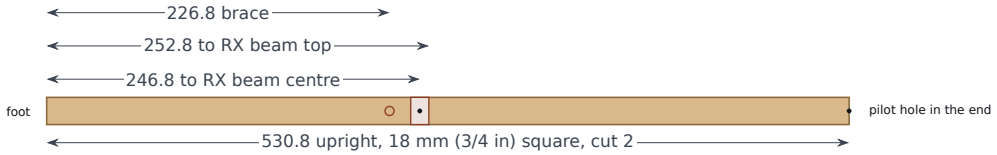


Figure 11: Upright, 1 : 5, drawn lying down, foot to the left. The shaded band is where the RX beam end butts in.

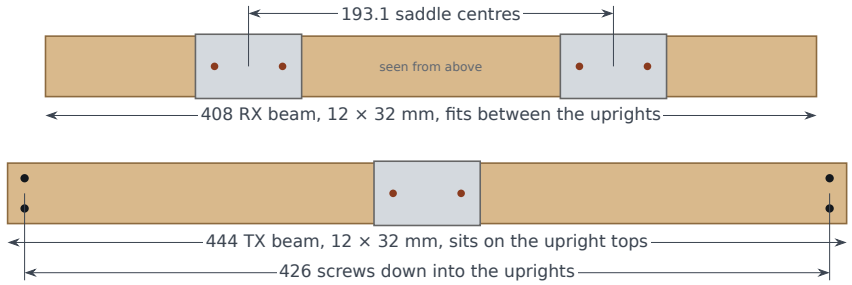


Figure 12: The two beams seen from above, 1 : 4. Grey are the saddle footprints, red the two screws that hold each saddle down.

4.3 The saddle clamp (full size)

One per horn. The pipe drops into the U, narrow wall down, and two M4 screws through the cheeks pinch it. Centre the saddle on the beam and slide the horn until the saddle is 90 mm behind the throat (where the flare meets the pipe) — that puts all three mouths in one line.

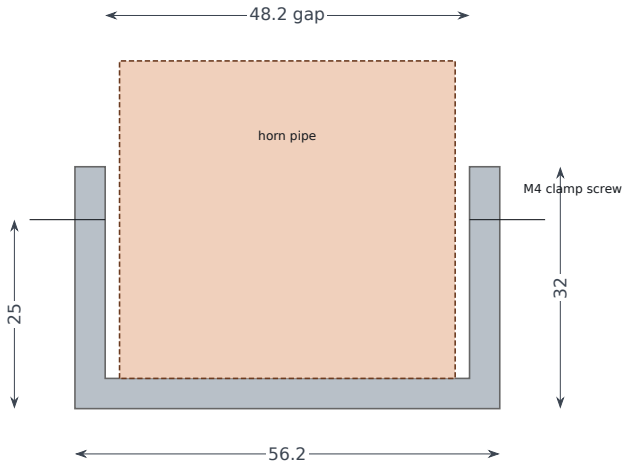
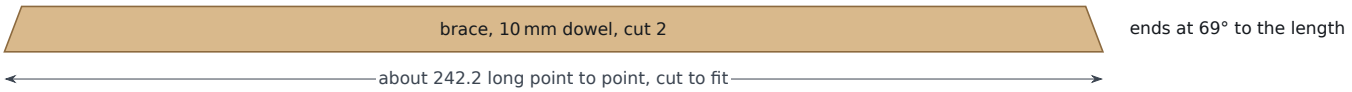


Figure 13: Saddle, front view at 1 : 1, 34 mm deep. 3-D print it in PETG at 40 % infill with the U opening upward, or bend it from 3 mm aluminium strip. Holes: 4.5 mm through both cheeks at the screw line; two 3.5 mm holes in the base for wood screws into the beam.

> **Do not crush the pipe**
>
> The inside of the pipe is what sets the horn’s frequency. Line the cheeks with a strip of rubber or foam tape and tighten the clamp screws only until the horn will not slide. A dented pipe is a detuned horn.



5 The RF chain

Seven bought modules, joined by SMA cables. There is nothing to make here; what matters is the order, the cable lengths, and that every connector is clean and tight. The modules go where the plan view on page 17 puts them: the transmit chain along the front edge, the receive chain behind it, so neither long cable has to cross the board.

5.1 Cable runs

run	from	to	cable	note
R1	ADF4351, RF OUT A	3 dB pad	20 cm	the other ADF output: leave its cap on, or a 50 Ω terminator
R2	3 dB pad	PA (SPF5189Z #1) IN	20 cm	or screw the pad straight onto the PA input
R3	PA OUT	splitter S (input)	20 cm	
R4	splitter port 1	TX horn	**1 m**	up the middle of the frame, tied every 10 cm
R5	splitter port 2	mixer LO	20 cm	the LO must be +10 to +13 dBm
R6	RX horn	band-pass IN	**60 cm**	down the middle of the frame, then right along the board. If 60 cm will not reach, 1 m costs only 0.5 dB
R7	band-pass OUT	LNA (SPF5189Z #2) IN	20 cm	
R8	LNA OUT	mixer RF	20 cm	
R9	mixer IF	breadboard J1	pigtail	SMA to two bare leads: centre to J1 pin 1, braid to pin 2

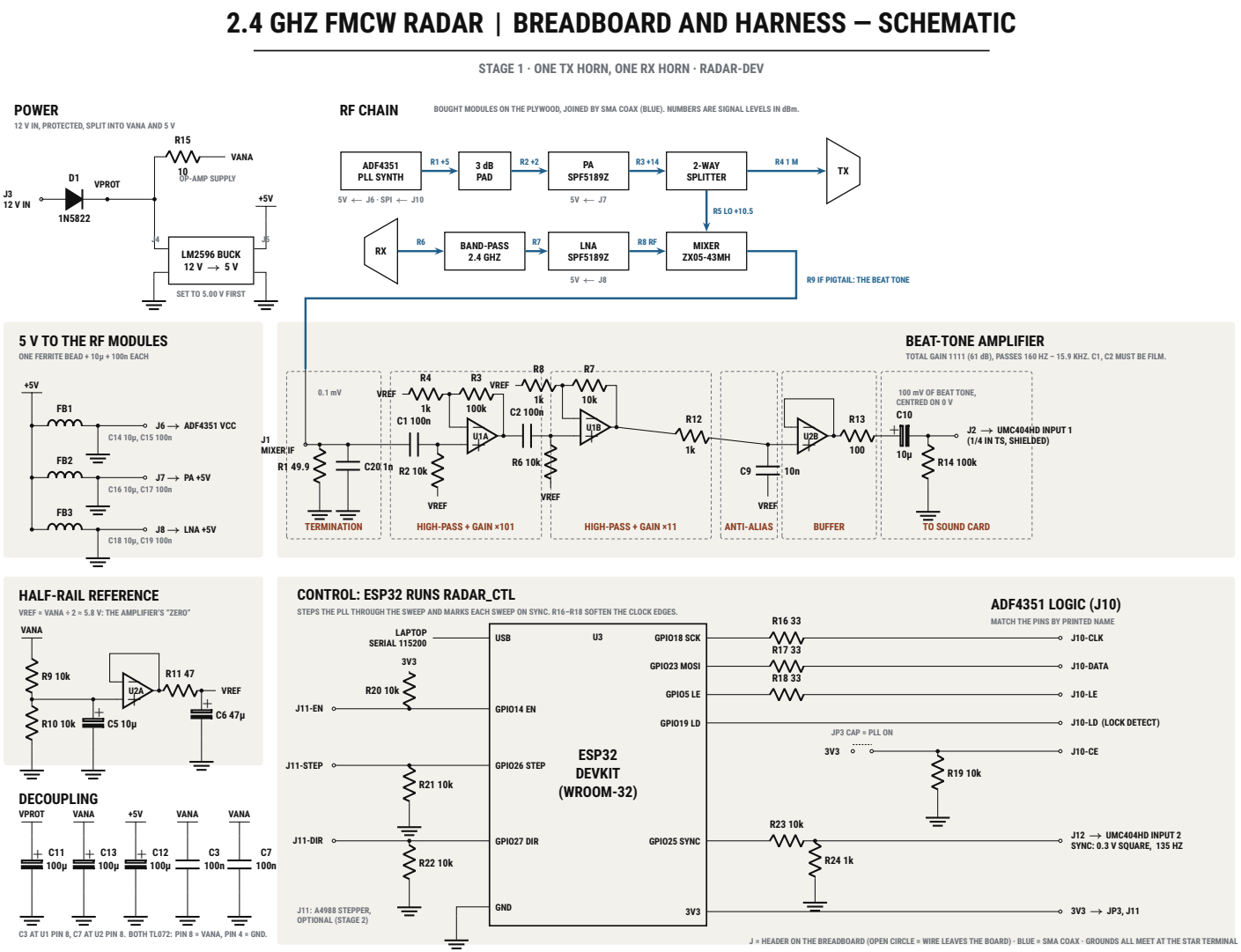
> **SMA connectors in one paragraph**
>
> Turn the nut, never the cable body. Finger tight, then an eighth of a turn with the spanner: that is all an SMA wants. Over-tightening splits the centre pin. Look into each connector before mating it; a bent pin or a fleck of solder in one ruins every measurement after it. Bend cables no tighter than a 2 cm radius.

5.2 Module power

The ADF4351, the PA and the LNA each take 5 V from the breadboard through their own ferrite bead and capacitors: J6, J7 and J8. Use red and black 22 AWG, twisted together, and keep them short. The mixer, splitter, pad and filter are passive and take no power.

module	supply	from	note
ADF4351 PLL	5 V, 120 mA	J6 (through FB1)	logic is 3.3 V; the ESP32 drives it directly through 33 Ω
PA, SPF5189Z #1	5 V, 90 mA	J7 (through FB2)	gets warm: that is normal
LNA, SPF5189Z #2	5 V, 90 mA	J8 (through FB3)	identical board to the PA
ESP32	USB	the laptop	also its serial link

6 The electronics, on one sheet



Also as a separate sheet: `build/schematic.pdf` (dark, for the screen) and `build/schematic-print.pdf`. Reference numbers (R1, C9, J2…) match the breadboard layout, the pick list and the assembly steps.

Reading the sheet

- **Top: where the signals come from.** The RF chain is bought modules on SMA coax (blue). The mixer's output, the beat tone, comes down the R9 pigtail into J1, the left end of the amplifier.
- **Middle: the beat-tone amplifier, left to right.** R1 is the mixer's 50 Ω load. C1 and C2 block DC and roll off below about 160 Hz, where the leakage and the room's clutter live; they *must* be film capacitors. U1A gains 101, U1B gains 11: 1111 in all, 61 dB. R12 and C9 cut off above 15.9 kHz. U2B buffers that filter so the sound card cannot load it, and C10 removes the DC so J2's output sits at 0 V.
- **VREF.** The op-amps run from a single 12 V supply, so instead of ground the signal swings around VREF, half of VANA (about 5.8 V), made by R9, R10 and the U2A buffer.
- **Power, left.** D1 blocks a backwards supply. VPROT feeds the buck converter and, through R15, the op-amps (VANA). The buck's 5 V goes to each RF module through its own ferrite bead and a 10 µ + 100 n pair, so their noise stays out of each other.
- **Bottom: control.** The ESP32 steps the PLL over SPI (R16–R18 soften the clock edges), reads lock detect back on GPIO19, and marks each sweep on SYNC, which the 10k/1k divider cuts to 0.3 V for the sound card's INPUT 2. JP3 enables the PLL; R19 holds it off otherwise. The stepper lines on the left are only for the optional turntable; their pull resistors keep the motor off at boot.

7 The breadboard

The circuits above, laid out on one 830-point breadboard. Signal flows left to right: the IF comes in at column 2, the amplified beat leaves at column 36; power enters bottom left, the module 5 V leaves at columns 38–45, the logic sits at the right. Part 2 builds it in seventeen steps, each with its own picture.

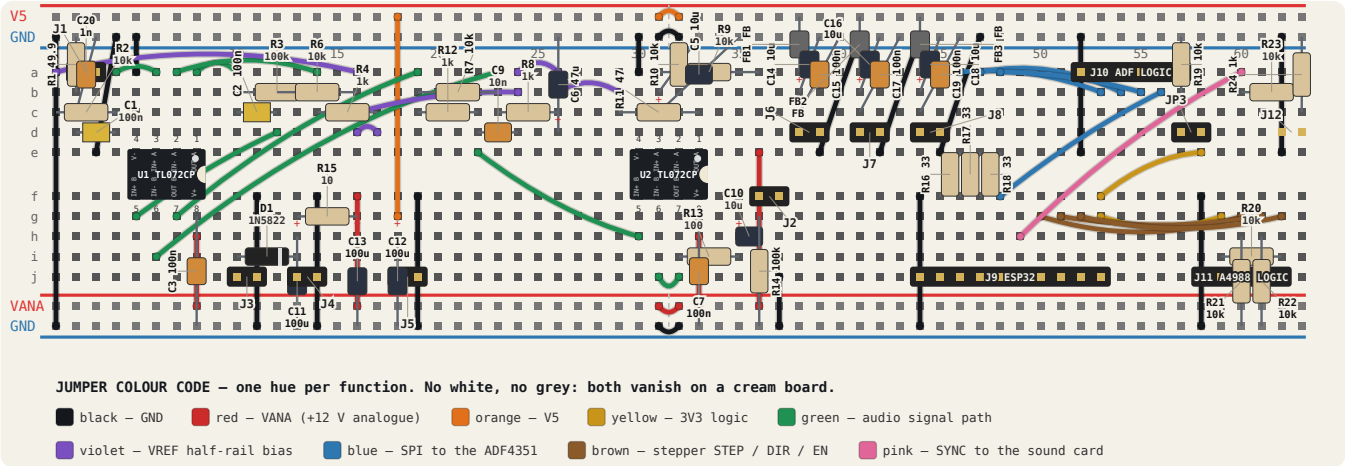


Figure 14: The finished breadboard. Columns 1–63 left to right; rows a–e on top, f–j below. Rails: top red 5 V, top blue GND, bottom red VANA (12 V analogue), bottom blue GND.

7.1 The jumper colour code

One colour per job, so a wire in the wrong place stands out. Nothing electrical depends on it; if your kit is short of a colour, substitute one and note it here.

	colour	carries	wires
■	black	GND	17
■	red	VANA, 12 V analogue	5
■	orange	5 V	2
■	yellow	3V3 logic	2
■	green	audio signal path	9
■	violet	VREF half-rail	5
■	blue	SPI to the ADF4351	4
■	brown	stepper STEP/DIR/EN	3
■	pink	SYNC	1

There is deliberately no white or grey wire: on a cream breadboard both disappear.

7.2 Pick list

Grouped by value, with how many to take from the drawer. **Bold** values set a gain, a cut-off or a termination: do not substitute those.

value	need	take	substitute?	refs
Resistors — ¼ W				
10k	10	14	**no** — gain, bias, and the half-rail divider. R9 and R10 want to match each other	R2, R6, R7, R9, R10, R19, R20, R21, R22, R23

value	need	take	substitute?	refs
1k	4	6	**no** — sets amplifier gain with the 100 k	R4, R8, R12, R24
33	3	5	22–68 Ω fine	R16, R17, R18
100k	2	3	**no** — sets gain with the 1 k	R3, R14
10	1	2	10–22 Ω fine	R15
100	1	2	68–150 Ω fine	R13
47	1	2	47–100 Ω fine	R11
49.9	1	2	**want 1 %** — this is the mixer IF termination. 47 Ω 5 % works if that is all there is	R1
Capacitors — ceramic				
100n	5	8	any; all of these are decoupling	C3, C7, C15, C17, C19
10n	1	2	**value matters** — sets the 15.9 kHz anti-alias corner with R12. C0G or film	C9
1n	1	2	any; C0G nicest	C20
Capacitors — film				
100n	2	3	**must be film, not ceramic** — signal path	C1, C2
Capacitors — electrolytic				
10u	5	7	any ≥ 16 V	C10, C5, C14, C16, C18
100u	3	4	**≥ 25 V** — two of these sit on the 12 V rail	C11, C13, C12
47u	1	2	any ≥ 16 V	C6
Diodes				
1N5822	1	2	any 1–3 A Schottky, ≥ 30 V	D1
Ferrite beads				
FB	3	4	any through-hole ferrite bead	FB1, FB2, FB3
Integrated circuits				
TL072CP	2	3	**DIP-8 package.** TL072, TL072C or TL082 all fit	U1, U2
Also				
4.7k	1	2	for R4 on first power-up only (gain 22 instead of 101)	R4
header strip	41 pins	2 strips	0.1 in male, snapped into 2-, 5-, 6- and 10-pin lengths	J1–J12, JP3

7.3 Connectors

Every wire that leaves the breadboard lands on a header. The holes column gives pin 1 first: on J12, pin 1 is the right-hand one.

ref	holes	pins	goes to
J1	a2, a3	1 IF, 2 GND	mixer IF via the R9 pigtail
J2	f36, f37	1 AUDIO_L, 2 GND	UMC404HD INPUT 1, 1/4 in TS cable
J3	j10, j11	1 +12 V, 2 GND	the 12 V supply
J4	j13, j14	1 VPROT, 2 GND	LM2596 IN+ / IN−
J5	j18, j19	1 5 V, 2 GND	LM2596 OUT+ / OUT−
J6	d38, d39	1 5 V, 2 GND	ADF4351 VCC / GND
J7	d41, d42	1 5 V, 2 GND	PA module + / −
J8	d44, d45	1 5 V, 2 GND	LNA module + / −

ref	holes	pins	goes to
J9	j44–j53	1 GND, 2 SCK, 3 MOSI, 4 LE, 5 LD, 6 SYNC, 7 STEP, 8 DIR, 9 EN, 10 3V3	ESP32: GND, D18, D23, D5, D19, D25, D26, D27, D14, 3V3
J10	a52–a57	1 GND, 2 CLK, 3 DATA, 4 LE, 5 LD, 6 CE	ADF4351 logic header, *by silkscreen name*: the boards differ in pin order
JP3	d57, d58	1 CE, 2 3V3	a jumper cap: on = PLL enabled
J11	j58–j62	1 GND, 2 3V3, 3 STEP, 4 DIR, 5 EN	A4988, optional, stage 2
J12	d63, d62	1 AUDIO_R, 2 GND	UMC404HD INPUT 2 (stage 1), 1/4 in TS cable